Octal Addition

base(r) = 8 ⇒ 0 to 7

$$\begin{array}{rcc} & 8^1 & 8^0 \\ & 1 & \\ & A_1 & \cancel{A_0}\ 5 \\ + & B_1 & \cancel{B_0}\ 4 \\ \hline & & 1 \end{array}$$

$\boxed{C_0 \le 7}$

Sum = C_0

Carry = 0

$C_0 > 7$ 8, 9 . . . . .

$$\boxed{C_0 = \underset{\text{carry}}{1} \times \underset{\text{base}}{8} + \underset{\text{sum}}{5}}$$

i)

$$\begin{array}{rccc} & 0 & 0 & \\ & 2 & 4 & 3 \\ + & 2 & 1 & 2 \\ \hline & 4 & 5 & 5 \end{array}$$

Ans

ii)

$$\begin{array}{rcccc} & 1 & 1 & 1 & \\ & & 5 & 6 & 7 \\ + & & 2 & 4 & 3 \\ \hline & 1 & 0 & 3 & 2 \end{array}$$

Ans

$$\begin{array}{r} 7 \\ +\ 7 \\ \hline 14 \end{array}$$

$14 = 1 \times 8 + 6$

$9 > 7$

$9 = 1 \times 8 + 1$

carry

9×8^0

$(8+1) \times 8^0$

$8 \times 8^0 + 1 \times 8^0$

$1 \times 8^1 + 1 \times 8^0$

carry

$10 > 7$

$10 = 1 \times 8 + 2$

$11 > 7$

$11 = 1 \times 8 + 3$

$8 > 7$

$8 = 1 \times 8 + 0$

H.W.

$$\begin{array}{rcccc} & 1 & 7 & 7 & 6 \\ + & & 3 & 4 & 5 \\ \hline & & & & \end{array}$$

2343

Octal Subtraction

Ex: 1

$$\begin{array}{r} \overset{6}{\cancel{7}}\ \overset{3}{\cancel{4}}\ 3 \\ -\ 5\ 6\ 4 \\ \hline 1\ 5\ 7 \end{array}$$

$8+3=11$ $8+3=11$

1 5 7 Ans

10
2
'8'

$7\times 8^2 + 4\times 8^1 + 3\times 8^0$

8×8^0

Ex: 2

$$\begin{array}{r} \overset{5}{\cancel{6}}\ \overset{1}{\cancel{2}}\ 4 \\ -\ 2\ 6\ 5 \\ \hline 3\ 3\ 7 \end{array}$$

$8+1=9$ $8+4=12$

3 3 7 Ans

H.w.

i)
$$\begin{array}{r} 5\ 1\ 1 \\ -\ 3\ 3 \\ \hline \end{array}$$

ii)
$$\begin{array}{r} 6\ 0\ 0\ 0 \\ -\ 7\ 7\ 7 \\ \hline \end{array}$$

1. 456
2. 5001

Octal Multiplication

$r = 8$

0 to 7

$14 = 1 \times 8 + 6$

$16 + 5 = 21$

$21 = 2 \times 8 + 5$

$17 = 2 \times 8 + 1$

$11 = 1 \times 8 + 3$

i)

$$\begin{array}{rrrr} & & 8^1 & 8^0 \\ & 1 & & \\ & & 7 & 2 \\ \times & & 1 & 2 \\ \hline 1 & 1 & 6 & 4 \\ & 7 & 2 & \times \\ \hline 1 & 1 & 0 & 4 \end{array}$$

Ans

$1 \times 8^1 \times 2 \times 8^0$

2×8^1

ii)

$$\begin{array}{rrrrr} & & 1 & 1 & \\ & 1 & 2 & 2 & \\ & & 3 & 5 & 7 \\ & \times & & 2 & 3 \\ \hline 1 & 1 & 3 & 1 & 5 \\ & 7 & 3 & 6 & \times \\ \hline 1 & 0 & 6 & 7 & 5 \end{array}$$

Ans

$8 = 1 \times 8 + 0$

$9 = 1 \times 8 + 1$

H.W

1) $\begin{array}{r} 63 \\ \times 24 \\ \hline \end{array}$

1774

2) $\begin{array}{r} 635 \\ \times 55 \\ \hline \end{array}$

44231